

A Calibrated Twist-Stress Toy-Model Hypothesis for CMB Temperature and Polarization in WRRRA

Wonsik Choi | Exploratory computational/model-development manuscript v0.5 | 10 October 2026

Abstract

We propose and calibrate an effective twist-stress toy model within Wonsik Reality Renderer Architecture (WRRRA), treating a dark gravitational component as a stress-energy phenotype. The present version adopts a pressureless CDM-equivalent closure and a power-law initial scalar spectrum, then calibrates its effective density, expansion scale, initial amplitude and tilt against Planck PR3 binned TT, TE and EE spectra. The ordinary-matter fraction is fixed at 0.050; radiation and massive neutrinos are explicitly included in flat density closure. A four-parameter bounded diagonal-error calibration yields $H_0=66.817$ km/s/Mpc, a total nonbaryonic matter fraction 0.272460, $A_s=2.106308 \times 10^{-9}$, and $n_s=0.962619$. The aggregate normalized residual RMS changes from 1.1062 to 1.0348. A deterministic withheld-bin diagnostic and an accuracy check are reported. The result is calibrated reproduction of familiar CMB structure under stated effective hypotheses, not a derivation of twist microphysics or a formal Planck likelihood test. Its intended contribution is a traceable model interface connecting the twist-stress interpretation and density accounting to actual CMB computation.

1. Hypothesis and research aim

The hypothesis is that a spatially distributed twist-stress component can possess an effective gravitational phenotype that reproduces observed CMB temperature and polarization structure. WRRRA organizes the interpretation through a finite source, common carrier and observable phenotype. This note implements the effective cosmological interface; it does not yet calculate the fluid closure or primordial spatial spectrum from finite-source arithmetic or common-carrier dynamics.

MCC 2.3 section 11.1 previously stipulated inheritance of conventional perturbation results under equal metric, stress-energy and initial conditions. Its section 11.2 demonstrated a 256×256 Zel'dovich growth toy, explicitly not a CMB transfer calculation. The twist-stress document assigns gravitational density to stress energy. Here we turn that prescription into an explicit, adjustable CMB toy-model profile and evaluate its numerical consequences [1,2].

Known observations and constants are legitimate calibration inputs. New numerical values are not required as the definition of originality. Explanatory contribution is assessed through the model's stated structure, transformation and implementation. A quantity fitted to the target data remains a calibration input; a later output at an unmeasured quantity would require its own fixed model specification and assessment.

2. Governing effective rules and accounting

Define $T(\text{total})=T(\text{ordinary})+T(\text{twist})+T(\text{radiation/neutrinos})+T(\text{background})$. Here ordinary means baryonic matter only; photons and neutrinos are assigned to their separate terms. This deliberately narrower bookkeeping definition prevents double counting relative to the broader ordinary-sector language in prior framework texts. The D component denotes the effective clustering twist sector, separately from standard massive neutrinos. Adopt $p_D=0$, ρ_D proportional to $a^{(-3)}$, rest-frame $c_{s,D}^2=0$, and zero anisotropic stress. These are constitutive hypotheses of this toy version, not claimed deductions from microscopic twist dynamics. In Newtonian gauge, $ds^2=a^2[-(1+2\psi)d\eta^2+(1-2\phi)dx^2]$, with conformal time η and $c=1$:

$$\delta_D' = -\theta_D + 3\phi'; \quad \theta_D' = -\mathcal{H}(\theta_D + k^2\psi).$$

We adopt scalar GR Einstein equations, standard baryon/photon/neutrino evolution and adiabatic initial conditions. CAMB's CDM slot implements the D-sector effective equations. The dimensionless primordial curvature spectrum is $P_R(k)=A_s (k/k^*)^{(n_s-1)}$, $k^*=0.05 \text{ Mpc}^{-1}$. A_s and n_s are calibrated effective initial-condition parameters associated with this twist interpretation. They are not derived from a twist action. An explicit transfer operator from the standard solver is adopted; no distinct twist solver is claimed.

For $h=H_0/100$ and fixed $\Omega_b=0.050$, $\omega_b=0.050h^2$. Let $\Omega_{D,\text{total}}$ include the standard massive-neutrino density. Then $\omega_D=\Omega_{D,\text{total}} h^2-\omega_v$, and $\Omega_D+\Omega_v=\Omega_{D,\text{total}}$. Photons and massless neutrinos supply Ω_r ; flat closure gives $\Omega_{\text{background}}=1-0.050-\Omega_{D,\text{total}}-\Omega_r$, with $w_{\text{background}}=-1$. This completes the late-time accounting through radiation domination in the chosen standard effective background.

WRR term	Computed variable / rule	Status
Twist gravitational phenotype	D in CAMB CDM slot	Interpretive assignment + adopted dust dynamics
Ordinary phenotype share	$\omega_b=0.050h^2$	Implemented coordinate restriction
Dark budget	$\omega_D=\Omega_{D,\text{total}} h^2-\omega_v$	Implemented density conversion
Return/background share	$\Omega_{\text{background}}=1-\Omega_b-\Omega_{D,\text{total}}-\Omega_r$	Implemented flat closure; adopted $w=-1$
Initial twist-related fluctuations	A_s and n_s in $P_R(k)$	Calibrated effective initial conditions
Finite source / common carrier / topology	No computed state-to-field or mode mapping	Not implemented in this module

This restricted effective module retains the twist-stress interpretation and density accounting. It adopts standard expansion, GR and flat isotropic transfer evolution. Earlier proposals about redshift as changing global twist, finite-volume modes, holonomy and galaxy-specific closures are left unconnected and not invoked in this calculation. The present result therefore does not establish consistency of those additional claims. With identical physical parameters, it is exactly the corresponding standard pressureless-component CMB family. The interface resolves the earlier unspecified baryon/neutrino split, radiation closure, primordial inputs and executable calibration recipe.

3. Calibration protocol

The adjustable parameters are H_0 , $\Omega_{D,\text{total}}$, A_s and n_s . Bounds are respectively [55,80] km/s/Mpc, [0.15,0.40], $[1.5,3.0] \times 10^{-9}$, and [0.90,1.05]. Initial values are 67.4, 0.268, 2.1×10^{-9} and 0.9649. $\tau=0.0544$, neutrino mass $\text{sum}=0.06$ eV, $N_{\text{eff}}=3.046$ and $T_{\text{CMB}}=2.7255$ K are fixed; standard recombination and BBN helium interpolation are adopted. This is a locally optimized effective calibration; global uniqueness is not claimed.

We use CAMB 2.0.4, $l_{\text{max}}=2500$, $\text{AccuracyBoost}=\text{lAccuracyBoost}=1.5$, $\text{lens_potential_accuracy}=0$ and NonLinear_none . Lensed spectra are computed from a linear-matter lensing potential; nonlinear corrections are omitted. This differs from the earlier exploratory run, so the before/after comparison is recomputed with one consistent prescription. A repeat at boosts 2.0 checks stability.

At each Planck bin representative multipole, interpolate $D_l=l(l+1)C_l/(2\pi)$. Define $r_i=(D_{\text{model}}-D_{\text{data}})/\sigma_i$, using the mean of the quoted asymmetric errors. Bounded scipy `least_squares` minimizes $\sum r_i^2$ across TT/TE/EE. No covariance, band windows or nuisance parameters are included. The objective and $R=\sqrt{\text{mean}(r_i^2)}$ are diagnostic measures, not a likelihood, p-value or acceptance threshold.

4. Results and evaluation chain

Evaluation step	Implementation and outcome
Inputs and status	Observed inputs: Planck PR3 spectra. Adopted effective physics: GR, recombination, dust closure and adiabatic spectrum. Fixed calibration assumptions: $\Omega_b=0.050$, τ and neutrino settings. Their evidential statuses differ.
WRRRA-specific transformation	Twist-stress gravitational phenotype -> separate neutrino/stress budget -> flat radiation-completed cosmology. The transfer operator and fluid equations are adopted, not source-derived.
Outputs	Calibrated lensed TT/TE/EE, peak positions, acoustic scale and sound horizon.
Falsification conditions	Implementation: bookkeeping/reproduction failure. Fixed effective profile: disagreement with proper observational tests. Future physical realization: failure to yield adopted stress/conservation rules. CMB cannot distinguish this twist interpretation from exactly equivalent CDM.

Parameter/result	Before	Calibrated
H0 [km/s/Mpc]	67.4	66.81651
$\Omega_{D,total}$	0.268	0.2724597
$A_s / 10^{(-9)}$	2.1	2.1063077
n_s	0.9649	0.9626194
$\Omega_{background}$	flat closure	0.6774593
Diagnostic R: TT	1.1268	0.9738
Diagnostic R: TE	1.0759	1.0979
Diagnostic R: EE	1.1098	1.0441
Aggregate R	1.1062	1.0348

Final TT integer-grid maxima: [220, 536, 813]. CAMB reports $100\theta^*=1.040656$, where θ^* is the sound horizon at last scattering divided by the angular-diameter distance in matching units; $r_{drag}=146.8844$ Mpc. These are conditional outputs of the calibrated effective model, not an independent BAO or BBN comparison.

Before fitting all bins, every fourth bin in the concatenated TT/TE/EE sequence was withheld (54 withheld, 161 training). Fitting training bins changed withheld R from 1.1342 to 1.1242. This is a diagnostic against purely training-specific improvement, not independent validation: adjacent published bins may be correlated. The final reported fit uses all bins.

The boost-2.0 repeat changes spectra by an RMS 0.00287 in units of the public bin errors. The high-accuracy aggregate R is 1.0347. Printed digits identify reproducible outputs, not parameter-confidence intervals. Optimizer status: `ftol` termination condition is satisfied

5. Calibration robustness and interpretation

Fixing $\Omega_b=0.050$ retains the framework's selected ordinary-phenotype fraction, while making ω_b depend on H_0 . It is a meaningful constraint on parameter space. As a sensitivity check, we instead fixed $\omega_b=0.0227138$ (the original physical baryon density) and refit the four other parameters, allowing Ω_b to change. This tests the restriction rather than silently replacing the main profile.

Check	H_0	$\Omega_{D,\text{total}}$	R_{all}
Main profile	66.8165	0.272460	1.03479
Alternative start and step	66.8125	0.272505	1.03479
Fixed physical baryon density	67.7721	0.261538	1.05721

The sensitivity fit gives $\Omega_b=0.049452$. The sensitivity changes the aggregate diagnostic from 1.0348 to approximately 1.0572 and changes the fitted density and expansion scale. Both remain of similar diagnostic magnitude, but the chosen baryon restriction materially affects the fitted parameters; neither fit yields a physical confidence interval. The alternative start was [70,0.25,2.2,0.98] with relative difference step 0.0005 instead of 0.001. These limited checks do not establish a global optimum or confidence region.

Rechecking the main endpoint yields objective $\Sigma r^2=230.219643$, scipy optimality=1.196602, termination: `xtol` termination condition is satisfied. A repeated training fit from the original initial values yields objective=162.881515, optimality=0.152074, termination: `xtol` termination condition is satisfied. These are solver stopping diagnostics, not statistical evidence for a unique physical parameter set. Full values and active bounds are supplied in `twist_robustness_results.json`.

An identical-input comparison against a direct conventional CAMB configuration gives maximum absolute D_I difference 0 μK^2 . This checks the wrapper mapping; it does not independently validate the twist interpretation. The boost test checks numerical accuracy within the stated linear-lensing prescription, not every nonlinear/lensing approximation.

The reported optimality values exceed the configured gradient tolerance $\text{gtol}=0.0001$. Termination by `ftol` or `xtol` is permitted, but these are stable calibration endpoints under the reported checks, not verified stationary optima.

The all-bin calibration improves TT and EE while TE becomes slightly worse. Withheld R improves only from 1.1342 to 1.1242: weak additional reassurance, not strong evidence of generalization. The main outcome is successful diagnostic calibration within a stated effective profile.

6. Temperature and polarization comparison

WRRR twist-stress toy: calibrated CMB reproduction (diagonal diagnostic)

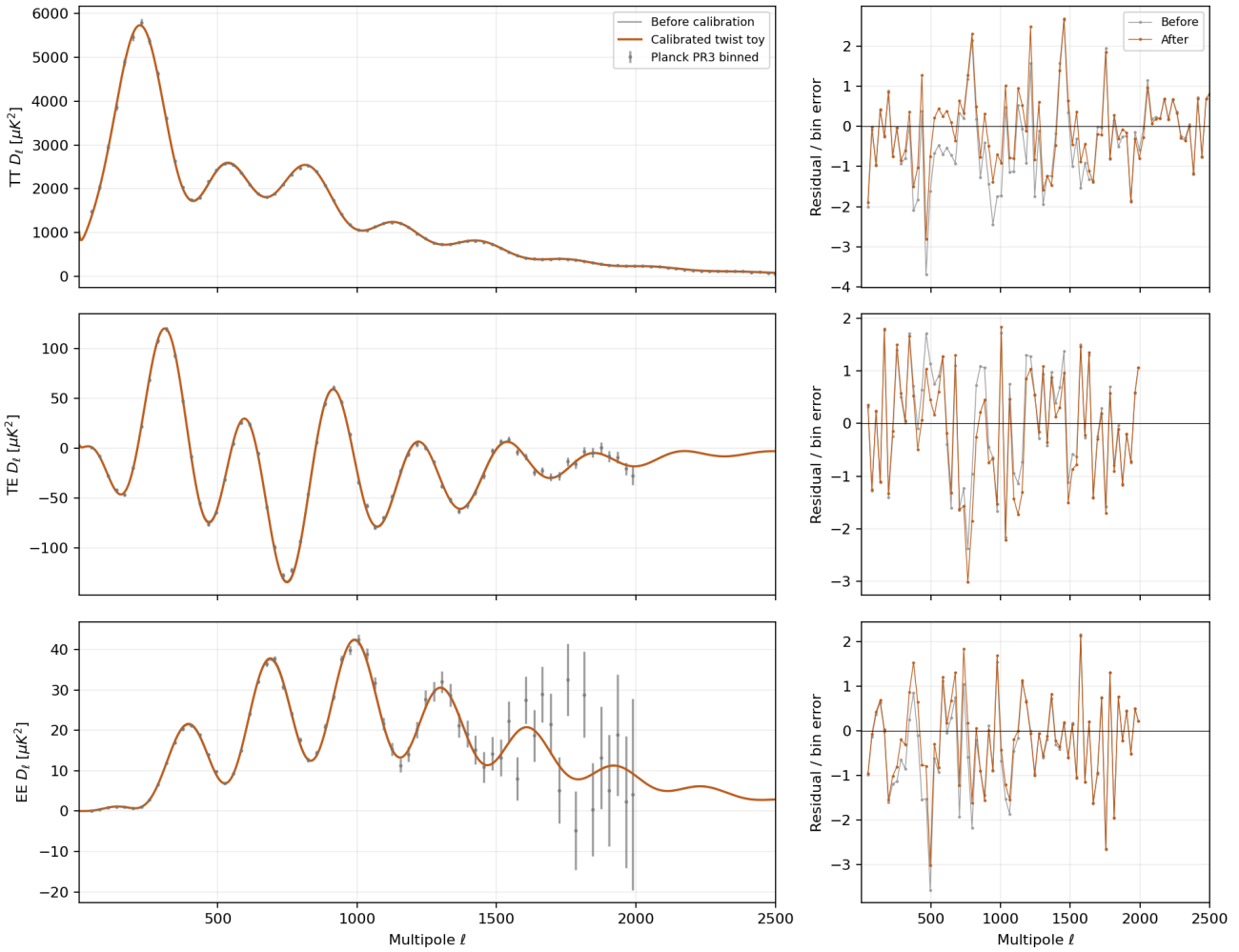


Figure 1. Planck PR3 binned spectra and the toy model before/after calibration. Right panels show diagnostic normalized residuals. Orange is the fitted effective twist profile. Matching data does not distinguish a particle interpretation from an exactly CDM-equivalent stress interpretation.

7. Interpretation and boundaries

The calibration improves the combined binned reproduction objective within this specified effective profile. Known-value reproduction is a consistency and explanatory result for the toy model. The actual completed calculation consists of budget mapping, calibrated initial conditions and adopted standard transfer evolution. It does not establish internal derivation of all those rules from WRRR's finite source.

CMB data cannot distinguish ontologies with identical effective evolution. A microscopic construction, alternative constitutive law or justified topology effect could enlarge the explanatory scope, but is not a prerequisite for defining and calibrating the present toy hypothesis. Such additional claims require actual computation before being described as integrated. Formal cosmological viability remains to be checked using covariance, band windows, nuisance parameters and the full likelihood. The calibration itself is not that test.

8. Reproducibility and further work

The submission supplement contains `calibrate_twist.py`, pinned CAMB requirements, raw Planck files, before/after spectra, boost-2.0 spectra, fit logs, `twist_calibration_results.json`, `check_twist_robustness.py` and `twist_robustness_results.json`. It also retains the prior v0.1 calculation and source excerpts. Execute `OMP_NUM_THREADS=2 python calibrate_twist.py` in the supplement directory after installing requirements; then `OMP_NUM_THREADS=2 python check_twist_robustness.py`. The checks read the calibration JSON, so run them second. Re-running performs training-only and all-bin fits; minor optimizer/solver numerical differences may occur. Tested environment: Python 3.12.14, NumPy 2.3.5, SciPy 1.17.0, CAMB 2.0.4. The complete environment record is `calibration_environment.json`. Data checksums are retained from the initial package.

Future work should test the effective profile with full CMB likelihoods and BAO/BBN observations, characterize parameter degeneracies, and decide which stress dynamics or finite-topology features WRRR computes internally. Any proposed extension must expose validated inputs, framework-specific transformations, outputs and falsification conditions on the same evidential standard applied to conventional models.

Author and AI assistance: Wonsik Choi. The numerical configuration, execution and writing used AI assistance. Neither this calibration nor its source package has been independently replicated by a human researcher. This working draft is not a journal acceptance or a registered public release; author declarations should be finalized before journal submission.

References

- [1] Choi, W. Minimal Computing Cosmology 2.3, integrated manuscript, 12 September 2026, sections 11.1-11.2; prior author manuscript, excerpt in supplement.
- [2] Choi, W. `MCC_DTU_Stress_Dark_Matter_v1.0_KO.md`; WRRR 3.0 v1.8, DOI 10.5281/zenodo.23233991. Snapshot: github.com/Wonsik-Choi-janefather/wrra-m-0.1, commit 03e11a92a952359a0c3908034d339a2e9eeca868.
- [3] Lewis, A., Challinor, A., Lasenby, A. Efficient computation of CMB anisotropies in closed FRW universes. *Astrophys. J.* 538, 473-476 (2000), [arXiv:astro-ph/9911177](https://arxiv.org/abs/astro-ph/9911177). CAMB documentation: camb.readthedocs.io/en/stable/.
- [4] Planck Collaboration. Planck 2018 results. VI. Cosmological parameters. *Astron. Astrophys.* 641, A6 (2020), [arXiv:1807.06209](https://arxiv.org/abs/1807.06209).
- [5] NASA/IPAC IRSA. Planck release 3 cosmoparams ancillary data, irsa.ipac.caltech.edu/data/Planck/release_3/ancillary-data/cosmoparams/. TT binned R3.01, TE/EE binned R3.02; accessed 10 October 2026.